

Conditional Concavity of p -Gromov–Wasserstein Energies

Why the quadratic exponent is exceptional, and a valid all-exponent extension

Gabriel Peyré

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Abstract

The usual conditional-concavity theorem for Gromov–Wasserstein transport uses the squared distortion $|d_X - d_Y|^2$ and same-sign conditionally definite base dissimilarities. It is tempting to replace the outer exponent 2 by an arbitrary $p \geq 1$. This note shows that such a replacement is false under the original assumptions: for every $p \neq 2$, explicit four-point CND metric spaces have a positive feasible curvature direction. Thus $p = 2$ is genuinely exceptional, not a technical limitation of the standard proof.

An exponent-uniform theorem nevertheless holds under a stronger geometric hypothesis. If all threshold kernels $\mathbf{1}_{\{d \geq r\}}$ of the two dissimilarities are conditionally negative definite, then every submodular scalar distortion with a nonnegative mixed measure—in particular $|d_X - d_Y|^p$ for $p \geq 1$ —has nonpositive curvature on zero-marginal directions. For genuine metrics, this threshold condition is equivalent to ultrametricity. A single layer-cake identity unifies this result with the classical quadratic theory. At $p = 2$, its scale weight is constant and integration recovers the usual CND tensor factorization; on ultrametric spaces, the same integrand is nonnegative at every pair of scales, which covers all $p \geq 1$. The finite result extends to arbitrary probability measures on compact ultrametric spaces. An extended catalogue connects this hypothesis to finite hierarchies, symbolic sequences, valued fields, profinite groups, rooted graphs and random tree boundaries. In the finite uniform case, concavity also guarantees a permutation optimizer.

1 Curvature along the coupling constraints

The GW objective is quadratic in the coupling for every fixed distortion kernel. Its convexity or concavity is therefore decided exactly by the sign of one quadratic form on directions that preserve both marginals.

Let X and Y be compact spaces, equipped with bounded symmetric dissimilarities $d_X : X^2 \rightarrow \mathbb{R}_+$ and $d_Y : Y^2 \rightarrow \mathbb{R}_+$. For $p \geq 1$, define

$$L_p((x, y), (x', y')) := |d_X(x, x') - d_Y(y, y')|^p. \quad (1)$$

For finite signed measures ξ, η on $Z := X \times Y$, set

$$\mathcal{B}_p(\xi, \eta) := \iint_{Z^2} L_p(z, z') d\xi(z) d\eta(z'), \quad \mathcal{Q}_p(\xi) := \mathcal{B}_p(\xi, \xi). \quad (2)$$

The p -GW problem minimizes $\mathcal{Q}_p(\pi)$ over $\pi \in \Pi(\alpha, \beta)$. Its feasible direction space is

$$\mathcal{T}_{X,Y} := \{\xi \in \mathcal{M}(X \times Y) : (P_X)_\# \xi = 0, (P_Y)_\# \xi = 0\}. \quad (3)$$

Proposition 1.1 (Exact feasible-curvature criterion). *For $\pi_0, \pi_1 \in \Pi(\alpha, \beta)$, write $\xi = \pi_1 - \pi_0$. Then*

$$\frac{d^2}{dt^2} \mathcal{Q}_p((1-t)\pi_0 + t\pi_1) = 2\mathcal{Q}_p(\xi). \quad (4)$$

Consequently, Q_p is concave on $\Pi(\alpha, \beta)$ whenever $Q_p(\xi) \leq 0$ for every $\xi \in \mathcal{T}_{X,Y}$. For finite spaces and strictly positive marginals, this condition is also necessary.

Proof. Expanding the quadratic form gives

$$Q_p(\pi_0 + t\xi) = Q_p(\pi_0) + 2t\mathcal{B}_p(\pi_0, \xi) + t^2Q_p(\xi),$$

which proves (4). Every difference of feasible couplings has zero marginals, so the sign condition implies concavity on each feasible segment. In the finite positive-marginal case, the product coupling lies in the relative interior of the transport polytope. Hence every zero-row- and zero-column-sum matrix is, after multiplication by a sufficiently small scalar of either sign, a feasible direction through that coupling. Concavity therefore forces its quadratic value to be nonpositive. $\square$

2 One multiscale identity, two concavity certificates

The quadratic and ultrametric theories are two levels of the same multiscale argument. The common object records the curvature contributed by every pair of distance thresholds. The quadratic loss averages those contributions with a constant weight, whereas a general power uses a non-flat scale measure that is singular on the diagonal when $p = 1$, and therefore requires finer control.

For a dissimilarity d and a threshold $r > 0$, define

$$H_r^d(x, x') := \mathbf{1}_{\{d(x, x') \geq r\}}. \quad (5)$$

For $\xi \in \mathcal{T}_{X,Y}$, its two-scale interaction is

$$I_\xi(r, s) := \iint H_r^{d_X}(x, x') H_s^{d_Y}(y, y') d\xi(x, y) d\xi(x', y'). \quad (6)$$

The same decomposition applies beyond power losses. Let D_X and D_Y bound the two dissimilarities.

Definition 2.1 (Submodular mixed-measure distortion). A continuous function $\ell : [0, D_X] \times [0, D_Y] \rightarrow \mathbb{R}$ is a *submodular mixed-measure distortion* if there is a finite positive Borel measure ω_ℓ on $(0, D_X] \times (0, D_Y]$ such that

$$\ell(a, b) = \ell(a, 0) + \ell(0, b) - \ell(0, 0) - \omega_\ell((0, a] \times (0, b]). \quad (7)$$

This is the measure-valued form of submodularity. If ℓ is twice continuously differentiable, the condition $\partial_{ab}^2 \ell \leq 0$ gives $d\omega_\ell(r, s) = -\partial_{ab}^2 \ell(r, s) dr ds$. The measure formulation also accommodates nonsmooth distortions concentrated on lower-dimensional sets.

Proposition 2.2 (Master multiscale curvature identity). For a finite signed measure $\xi \in \mathcal{T}_{X,Y}$, define

$$Q_\ell(\xi) := \iint \ell(d_X(x, x'), d_Y(y, y')) d\xi(x, y) d\xi(x', y'). \quad (8)$$

If ℓ satisfies [definition 2.1](#), then

$$Q_\ell(\xi) = - \int I_\xi(r, s) d\omega_\ell(r, s). \quad (9)$$

In particular, for $\ell_p(a, b) = |a - b|^p$,

$$Q_1(\xi) = -2 \int_0^\infty I_\xi(r, r) dr, \quad (10)$$

$$Q_p(\xi) = -p(p-1) \int_0^\infty \int_0^\infty |r-s|^{p-2} I_\xi(r, s) dr ds, \quad p > 1. \quad (11)$$

Proof. Insert (7) into (8). Terms depending only on d_X , only on d_Y , or on neither vanish because ξ has zero marginals. Moreover,

$$\omega_\ell((0, a] \times (0, b]) = \int \mathbf{1}_{\{a \geq r\}} \mathbf{1}_{\{b \geq s\}} d\omega_\ell(r, s).$$

Fubini's theorem therefore gives (9).

For $p = 1$, use $a + b - |a - b| = 2 \int_0^\infty \mathbf{1}_{\{a \geq r\}} \mathbf{1}_{\{b \geq r\}} dr$. For $p > 1$, two integrations of the mixed derivative give

$$a^p + b^p - |a - b|^p = p(p-1) \int_0^a \int_0^b |r - s|^{p-2} dr ds.$$

The kernel $|r - s|^{p-2}$ is locally integrable exactly for $p > 1$, which proves both power formulas. $\square$

The quadratic exponent integrates the scales. Setting $p = 2$ in (11) makes the scale weight equal to one. Since $\int_0^\infty H_r^d(x, x') dr = d(x, x')$, Fubini gives

$$Q_2(\xi) = -2 \int_0^\infty \int_0^\infty I_\xi(r, s) dr ds = -2 \iint d_X(x, x') d_Y(y, y') d\xi(x, y) d\xi(x', y'). \quad (12)$$

Thus the familiar quadratic factorization is not a separate theory: it is the flat-weight member of the multiscale identity.

Corollary 2.3 (Integrated quadratic certificate). *If d_X and d_Y are both conditionally negative definite (CND), or both conditionally positive definite (CPD), then $Q_2(\xi) \leq 0$ for every $\xi \in \mathcal{T}_{X,Y}$. Hence squared GW is concave on each transport polytope.*

Proof. Center each dissimilarity at an arbitrary base point. Additive one-variable terms disappear against the zero marginals of ξ . In the CND case, the negatives of the two centered kernels are positive semidefinite; in the CPD case, the centered kernels themselves are positive semidefinite. Their Kronecker-product quadratic form is therefore nonnegative in both cases. Equation (12) gives the result. $\square$

This is the mechanism behind conditional concavity in graph matching and squared GW (Maron and Lipman, 2018; Séjourné et al., 2021). Crucially, only the *integrated* kernels d_X and d_Y need to have the appropriate sign; the individual threshold interactions $I_\xi(r, s)$ may change sign.

The word *squared* here refers to the outer exponent in (1). It should not be confused with an inner power in the base dissimilarities. For example, Schoenberg's theorem implies that $\|x - x'\|^q$ is CND for $0 < q \leq 2$ (Schoenberg, 1938); each such inner exponent is covered by (12), but the outer exponent remains exactly 2. Recent Hilbert-embedding formulations likewise broaden the base costs while retaining a squared outer discrepancy (Houry et al., 2026).

Ultrametric geometry controls every scale separately. Formula (9) reveals a stronger sufficient condition: if $I_\xi(r, s) \geq 0$ for every feasible direction ξ and every pair of thresholds, then every submodular mixed-measure distortion is conditionally concave. Threshold-CND dissimilarities provide exactly this pointwise certificate. For genuine metrics, the next sections show that this condition is equivalent to ultrametricity. Consequently, the ultrametric theory contains the quadratic result as its $p = 2$ member, but strengthens its hypothesis in order to cover all the other scale measures as well.

3 The CND hypothesis does not extend beyond $p = 2$

The collapse of the multiscale formula to the product factorization (12) is specific to its constant $p = 2$ weight. For other exponents, [proposition 2.2](#) retains scale-dependent information whose sign is not controlled by ordinary CND. The failure already occurs for four-point path, tree and Hamming metrics, so it persists even when the base kernels are genuine CND distances rather than arbitrary CND dissimilarities.

Theorem 3.1 (The original hypothesis is specific to $p = 2$). *For every $p \in [1, \infty) \setminus \{2\}$, there are four-point metric spaces (X, d_X) and (Y, d_Y) whose distance kernels are CND, and a nonzero matrix $\Xi \in \mathcal{T}_{X,Y}$, such that $Q_p(\Xi) > 0$. Hence same-sign conditional definiteness of genuine metric distances does not imply conditional concavity for any outer exponent other than 2.*

Proof. We use only finite ℓ^1 and tree metrics. They are CND: in one dimension,

$$|s - t| = \int_{\mathbb{R}} |\mathbf{1}_{\{s > r\}} - \mathbf{1}_{\{t > r\}}|^2 dr,$$

so, for $\sum_i a_i = 0$, its quadratic form equals $-2 \int_{\mathbb{R}} |\sum_i a_i \mathbf{1}_{\{s_i > r\}}|^2 dr \leq 0$. Finite sums preserve conditional negative definiteness, proving the claim for ℓ^1 ; tree metrics are sums of the same cut metrics.

First let $1 \leq p < 2$. Take the line metric on the labeled points $(0, 3, 2, 1)$ and the Hamming metric on the labeled square $((0, 0), (1, 1), (1, 0), (0, 1))$. Their distance matrices are

$$D_X = \begin{pmatrix} 0 & 3 & 2 & 1 \\ 3 & 0 & 1 & 2 \\ 2 & 1 & 0 & 1 \\ 1 & 2 & 1 & 0 \end{pmatrix}, \quad D_Y = \begin{pmatrix} 0 & 2 & 1 & 1 \\ 2 & 0 & 1 & 1 \\ 1 & 1 & 0 & 2 \\ 1 & 1 & 2 & 0 \end{pmatrix}. \quad (13)$$

Set $u = (1, -1, -1, 1)^\top$, $v = (1, 1, -1, -1)^\top$ and $\Xi_{<} = uv^\top$. Since both vectors sum to zero, $\Xi_{<}$ has zero row and column sums. A direct collection of the distance differences in (13) gives

$$Q_p(\Xi_{<}) = 8(2^{p+1} - 3^p + 1) > 0. \quad (14)$$

Indeed, $\varphi(p) = (3/2)^p - 2^{-p}$ is strictly increasing and $\varphi(2) = 2$. Hence $p < 2$ implies $3^p - 1 < 2^{p+1}$, proving the strict inequality in (14). This includes $p = 1$.

Now let $p > 2$. Keep D_Y from (13), but let X be the unit-edge star with three leaves and its center as the fourth point:

$$D_X = \begin{pmatrix} 0 & 2 & 2 & 1 \\ 2 & 0 & 2 & 1 \\ 2 & 2 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}. \quad (15)$$

With $\tilde{u} = (1, 0, 0, -1)^\top$, use the feasible direction $\Xi_{>} = \tilde{u}v^\top$. Direct summation now yields

$$Q_p(\Xi_{>}) = 8(2^p - 4) > 0, \quad (16)$$

which is strictly positive exactly when $p > 2$. $\square$

Remark 3.2 (What the counterexamples rule out). The theorem does not say that every $p \neq 2$ GW objective is nonconcave. It says that CND or CPD base dissimilarities alone no longer certify concavity. Any valid extension must impose additional structure on the base geometry or directly on the distortion kernel. It also shows why replacing the tensor factorization in the $p = 2$ proof by an appeal to Schoenberg's theorem would be invalid: Schoenberg controls powers of a single distance kernel, whereas $|d_X - d_Y|^p$ couples two distance values through a nonlinear difference.

4 Pointwise threshold positivity and ultrametrics

The master identity isolates the extra structure needed away from the quadratic exponent: positivity must hold before the distance scales are averaged. Requiring conditional negative type at every threshold supplies a clean geometric certificate.

Definition 4.1 (Threshold negative type). A dissimilarity is *threshold-CND* if every kernel H_r^d defined in (5) is CND. On an infinite space, this means that the CND inequality holds on every finite subset.

Threshold-CND implies CND because $d(x, x') = \int_0^\infty H_r^d(x, x') dr$. For genuine metrics, the stronger condition has an exact geometric meaning.

Proposition 4.2 (Threshold negative type characterizes ultrametrics). *A metric d is threshold-CND if and only if it is an ultrametric, namely*

$$d(x, z) \leq \max\{d(x, y), d(y, z)\} \quad \text{for all } x, y, z. \quad (17)$$

Proof. Suppose first that d is ultrametric. For each $r > 0$, the relation $x \sim_r x'$ if $d(x, x') < r$ is an equivalence relation. On any finite collection $(x_i)_i$, let $\mathcal{P}_r$ denote the resulting partition. For coefficients satisfying $\sum_i a_i = 0$,

$$\sum_{i,j} a_i a_j H_r^d(x_i, x_j) = - \sum_{C \in \mathcal{P}_r} \left(\sum_{i: x_i \in C} a_i \right)^2 \leq 0.$$

Thus every H_r^d is CND.

Conversely, if (17) fails, there are x, y, z such that $d(x, z) > \max\{d(x, y), d(y, z)\}$. Choose r strictly above the latter maximum and no larger than $d(x, z)$. The threshold kernel vanishes on (x, y) and (y, z) , but equals one on (x, z) . Its quadratic form at the zero-sum coefficients $(1, -2, 1)$ is therefore $2 > 0$, contradicting conditional negative definiteness. $\square$

Thus threshold-CND is much stronger than ordinary negative type among metrics: it selects precisely ultrametric geometry. The line and tree metrics used in [theorem 3.1](#) are CND but not ultrametric.

Theorem 4.3 (Submodular distortions under threshold negative type). *Let X and Y be finite spaces whose dissimilarities are threshold-CND, and let ℓ satisfy [definition 2.1](#). Then $Q_\ell(\Xi) \leq 0$ for every $\Xi \in \mathcal{T}_{X,Y}$. Consequently, the associated quadratic transport objective is concave on every transport polytope over the two spaces.*

Proof. Since both threshold kernels are CND, centering their negatives at arbitrary base points produces positive semidefinite matrices. All centering terms disappear because Ξ has zero row and column sums. Hence $I_\Xi(r, s)$ is the quadratic form of a Kronecker product of two positive semidefinite matrices, and is nonnegative; the two minus signs cancel. The master identity (9) now gives $Q_\ell(\Xi) \leq 0$. $\square$

Corollary 4.4 (All power distortions). *Under the assumptions of [theorem 4.3](#), the choice $\ell_p(a, b) = |a - b|^p$ gives $Q_p(\Xi) \leq 0$ for every $p \geq 1$. In particular, $p = 2$ is included as the constant-weight specialization of (11).*

Proof. Apply [theorem 4.3](#) to the power-loss representations in [proposition 2.2](#). $\square$

Corollary 4.5 (Sparse and permutation minimizers). *Let X and Y be finite ultrametric spaces with strictly positive marginals supported on n and m points. For every $p \geq 1$, the p -GW problem admits an optimal coupling supported on at most $n + m - 1$ pairs. If $n = m$ and both marginals are uniform, it admits a permutation coupling as an optimizer.*

Proof. By [proposition 4.2](#) and [corollary 4.4](#), the objective is concave on the transport polytope. Write any minimizer as a convex combination of extreme points. Concavity shows that its value is no smaller than the corresponding convex combination of their values; optimality forces every extreme point used with positive weight to be optimal. An extreme transport plan has at most $n + m - 1$ positive entries. For equal uniform marginals, the Birkhoff–von Neumann theorem identifies the extreme points with scaled permutation matrices. $\square$

Why this is the right strengthening. The two theories are now nested rather than disconnected. Ordinary CND controls the scale average $d = \int H_r^d dr$, which suffices when $p = 2$ because the weight in (11) is constant. Threshold-CND controls $I_\xi(r, s)$ pointwise, so every nonnegative mixed measure may be used; for metrics, this is precisely the ultrametric hypothesis. The latter therefore recovers the quadratic theorem and extends it to all $p \geq 1$. It is sufficient, but not claimed to be necessary for concavity at one fixed exponent and one fixed pair of spaces: the exact finite criterion remains [proposition 1.1](#).

5 A hierarchy of ultrametric examples

Ultrametrics range from the constant distance on a finite set to spaces of infinite trees, valued fields and locally convergent graphs. Their common mechanism is not an exotic triangle inequality but a nested family of partitions: two points are close when they remain in the same cluster down to a fine scale. This section builds a catalogue from elementary examples to more structural ones and records compactness, because compact ultrametric spaces enter directly into [theorem 6.1](#).

Nested partitions are the basic construction. The finite case already contains the complete geometry. It also explains why dendrograms and ultrametrics are equivalent representations of the same hierarchy ([Carlsson and Mémoli, 2010](#)).

Proposition 5.1 (Finite hierarchies and ultrametrics). *Let $\mathcal{P}_0, \dots, \mathcal{P}_L$ be partitions of a finite set X , where $\mathcal{P}_0$ is the partition into singletons, $\mathcal{P}_L = \{X\}$, and each $\mathcal{P}_\ell$ refines $\mathcal{P}_{\ell+1}$. Fix heights $0 = h_0 < h_1 < \dots < h_L$. For $x \neq x'$, set*

$$m(x, x') := \min\{\ell : x, x' \text{ lie in the same block of } \mathcal{P}_\ell\}, \quad d(x, x') := h_{m(x, x')}, \quad (18)$$

and set $d(x, x) = 0$. Then d is an ultrametric. Conversely, every finite ultrametric is obtained in this way.

Proof. If x, y and y, z belong to common blocks by level $\ell = \max\{m(x, y), m(y, z)\}$, those blocks intersect at y and hence are the same block. Thus $m(x, z) \leq \ell$, which gives $d(x, z) \leq \max\{d(x, y), d(y, z)\}$.

Conversely, let $0 = h_0 < h_1 < \dots < h_L$ be the distinct values of a finite ultrametric. At level ℓ , declare $x \sim_\ell x'$ when $d(x, x') \leq h_\ell$. The strong triangle inequality makes this relation transitive, so its classes form a partition $\mathcal{P}_\ell$. These partitions are nested, and the first level at which x and x' merge has height exactly $d(x, x')$. $\square$

Example 5.2 (The constant discrete ultrametric). On any set X , the metric

$$d(x, x') = R \mathbf{1}_{\{x \neq x'\}}, \quad R > 0,$$

is ultrametric. Every triangle is either degenerate or equilateral. On a finite set, this is (18) with only the singleton partition and the one-block partition. It is the simplest example, but also the least informative one: there is only one nontrivial distance threshold.

Example 5.3 (A two-scale block model). Let $0 < r < R$, split $\{1, 2, 3, 4\}$ into the two blocks $\{1, 2\}$ and $\{3, 4\}$, and use distance r inside a block and R across blocks. The distance matrix is

$$\begin{pmatrix} 0 & r & R & R \\ r & 0 & R & R \\ R & R & 0 & r \\ R & R & r & 0 \end{pmatrix}.$$

At thresholds up to r , every point is isolated; between r and R , the two prescribed blocks are visible; above R , everything has merged. Replacing the two blocks by an arbitrary nested collection gives a finite hierarchical block model.

Example 5.4 (Dendrograms and weighted rooted trees). Let the points be the leaves of a rooted tree. Give every vertex a height, with leaves at height zero and heights strictly increasing toward the root. If $x \wedge x'$ denotes the lowest common ancestor of two leaves, then

$$d(x, x') = h(x \wedge x') \quad (19)$$

is ultrametric: among three leaves, the two largest merge heights coincide. Conversely, [proposition 5.1](#) turns every finite ultrametric into such a dendrogram. This is the geometry produced by hierarchical clustering and used for taxonomies and phylogenetic representations. It is important not to confuse (19) with the shortest-path metric on all vertices of a tree; the latter is generally a tree metric, but not an ultrametric.

Prefix spaces turn hierarchy into coordinates. Words and sequences provide the canonical infinite-scale model: distance is determined by the first coordinate at which two objects disagree.

Example 5.5 (Finite words, Cantor space and Baire space). Let $\mathcal{A}$ be an alphabet, let $0 < \vartheta < 1$, and consider sequences $x = (x_k)_{k \geq 1}$ in $\mathcal{A}^{\mathbb{N}}$. If $x \neq x'$, define the first disagreement $N(x, x') = \min\{k \geq 1 : x_k \neq x'_k\}$ and

$$d_{\vartheta}(x, x') = \vartheta^{N(x, x')-1}, \quad d_{\vartheta}(x, x) = 0. \quad (20)$$

If two pairs agree through the first m symbols, then so does the remaining pair; hence (20) is ultrametric. Its balls are cylinder sets obtained by fixing an initial word. Fixed-length words inherit finite ultrametrics. If $\mathcal{A}$ is finite, the infinite sequence space is compact and is usually called a Cantor space; if $\mathcal{A} = \mathbb{N}$, one obtains the complete but noncompact Baire space. Closed subshifts and spaces of admissible symbolic trajectories are ultrametric subspaces of the same form.

Example 5.6 (The ternary Cantor set). The coding map

$$\iota(\omega) = \sum_{k \geq 1} 2\omega_k 3^{-k}, \quad \omega \in \{0, 1\}^{\mathbb{N}},$$

identifies binary sequences with the middle-third Cantor set. Transporting (20) with $\vartheta = 1/3$ through ι gives an ultrametric d_C on that fractal. If N is the first differing digit, then

$$\frac{1}{3} d_C(\omega, \omega') \leq |\iota(\omega) - \iota(\omega')| \leq d_C(\omega, \omega'). \quad (21)$$

Thus the familiar Euclidean geometry of the Cantor set is bi-Lipschitz equivalent to an ultrametric geometry, although the restricted Euclidean distance itself is not ultrametric. This example is a useful warning that ultrametricity depends on the chosen distance, not only on the topology.

Valuations produce arithmetic ultrametrics. The strong triangle inequality is the defining feature of non-Archimedean absolute values. Congruence at increasing precision plays exactly the role of agreement along an increasingly long prefix.

Example 5.7 (Finite congruence spaces and the p -adic integers). Fix a prime p . For distinct residues $a, b \in \mathbb{Z}/p^L\mathbb{Z}$, let

$$n(a, b) = \max\{0 \leq k < L : a \equiv b \pmod{p^k}\}, \quad d_p(a, b) = p^{-n(a, b)},$$

and set $d_p(a, a) = 0$. Congruence modulo p^k gives the nested partitions, so this is a finite ultrametric. Passing to the inverse limit gives the compact ring of p -adic integers $\mathbb{Z}_p$, with

$$d_p(x, y) = |x - y|_p = p^{-v_p(x-y)}. \quad (22)$$

The valuation inequality $v_p(x + y) \geq \min\{v_p(x), v_p(y)\}$ is precisely the ultrametric inequality. Balls are congruence classes modulo p^k . The field $\mathbb{Q}_p$ has the same local geometry but is not compact; compact subsets such as $\mathbb{Z}_p$, or probability measures supported on compact balls, are therefore the direct setting for the compact GW statement.

Example 5.8 (Formal series and non-Archimedean vector spaces). For a finite field $\mathbb{F}_q$, let $\text{ord}_t(f)$ be the lowest nonzero exponent in a Laurent series $f \in \mathbb{F}_q((t))$. Then

$$d(f, g) = q^{-\text{ord}_t(f-g)}$$

is ultrametric, and the unit ball $\mathbb{F}_q[[t]]$ is compact. More generally, if $\mathbb{K}$ is any non-Archimedean valued field, then $d(x, y) = \|x - y\|_\infty$ is ultrametric on $\mathbb{K}^m$, as well as on sequence spaces such as $c_0(\mathbb{N}, \mathbb{K})$. Hence ultrametric spaces are not confined to trees: they also arise as finite- and infinite-dimensional algebraic state spaces.

Closure rules generate more elaborate spaces. Several constructions preserve the strong triangle inequality and can be used to build new GW input spaces from the preceding examples.

Proposition 5.9 (Elementary closure properties). *Let (X, d) and (X_j, d_j) , $1 \leq j \leq m$, be ultrametric spaces.*

1. *Every subset of X , with the restricted distance, is ultrametric.*
2. *If $\varphi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is nondecreasing, $\varphi(0) = 0$, and $\varphi(r) > 0$ for $r > 0$, then $\varphi \circ d$ is ultrametric. In particular, powers, truncations and bounded transforms such as $d/(1+d)$ preserve ultrametricity.*
3. *The max product*

$$d_{\max}(x, x') = \max_{1 \leq j \leq m} d_j(x_j, x'_j)$$

is ultrametric on $\prod_j X_j$.

4. *On the nonempty compact subsets $A, B \subset X$, the Hausdorff distance*

$$d_H(A, B) = \max \left\{ \sup_{x \in A} \inf_{y \in B} d(x, y), \sup_{y \in B} \inf_{x \in A} d(x, y) \right\} \quad (23)$$

is ultrametric. If X is compact, then its Hausdorff hyperspace is compact as well.

Proof. Restriction is immediate. Monotonicity gives $\varphi(d(x, z)) \leq \varphi(\max\{d(x, y), d(y, z)\})$, and a nondecreasing function commutes with the maximum of two scalars. Taking the maximum of the componentwise strong triangle inequalities proves the product claim.

For the Hausdorff claim, fix $r > \max\{d_H(A, B), d_H(B, C)\}$. Every $x \in A$ lies within r of some $y \in B$, which lies within r of some $z \in C$. Ultrametricity gives $d(x, z) < r$. The same argument in the reverse direction proves $d_H(A, C) \leq \max\{d_H(A, B), d_H(B, C)\}$. Compactness of the hyperspace follows by approximating compact sets with subsets of a common finite ε -net and using completeness. $\square$

Example 5.10 (Profinite groups and inverse limits). Let $G = G_0 \supset G_1 \supset \dots$ be finite-index normal subgroups with $\bigcap_k G_k = \{e\}$, and consider the inverse-limit completion. For $0 < \vartheta < 1$, define

$$N(g, h) = \sup\{k : g^{-1}h \in G_k\}, \quad d(g, h) = \vartheta^{N(g, h)},$$

with $\vartheta^\infty = 0$. Two elements are close when they agree in many finite quotients. Since agreement through level k is an equivalence relation, this is a compact ultrametric group; normality makes the distance bi-invariant. The additive group of $\mathbb{Z}_p$ is recovered from $G_k = p^k \mathbb{Z}_p$, while products of finite groups give prefix metrics on increasingly refined discrete coordinates.

Example 5.11 (The local topology on rooted graphs). Let $\mathcal{G}_D$ be the isomorphism classes of connected rooted simple graphs with degree at most D . For rooted graphs (G, o) and (G', o') , let $R(G, G')$ be the largest radius for which the rooted radius- R neighborhoods are isomorphic, and set

$$d_{\text{loc}}(G, G') = 2^{-R(G, G')}, \quad 2^{-\infty} = 0. \quad (24)$$

Agreement of two rooted balls through radius R is transitive, so (24) is ultrametric. Bounded degree leaves only finitely many rooted ball types at each radius; a diagonal argument then shows that $\mathcal{G}_D$ is compact. Weak convergence of probability laws on this space is the local weak, or Benjamini–Schramm, convergence used for large random graphs (Benjamini and Schramm, 2001).

Example 5.12 (Compact-set-valued data). Starting from any compact example above, one may take its nonempty compact subsets and compare them with (23). Iterating the construction gives spaces of hierarchical set-valued objects, all still compact and ultrametric. For a finite dendrogram, a set is observed at resolution r only through the collection of level- r clusters it meets; two sets are within r exactly when these collections agree. This makes the threshold partition behind the induced Hausdorff ultrametric explicit.

Example 5.13 (Random tree boundaries and hierarchical cascades). Let $\mathcal{T}$ be an infinite locally finite rooted tree and let $\partial\mathcal{T}$ be its infinite rays. Choose radii $r_0 > r_1 > \dots \downarrow 0$. If $x \wedge y$ is the deepest common vertex of two rays, then

$$d(x, y) = r_{|x \wedge y|} \quad (x \neq y), \quad d(x, x) = 0,$$

is an ultrametric, and $\partial\mathcal{T}$ is compact. Assigning random child weights that sum to one at each vertex produces a random probability measure whose cylinder mass is the product of the weights along its branch. The geometry remains ultrametric while the measure becomes highly inhomogeneous. Such random hierarchical measures are the elementary template behind multiplicative cascades and the ultrametric organization of states in spin-glass models; Ruelle probability cascades provide a canonical infinite-level realization (Rammal et al., 1986; Ruelle, 1987).

Remark 5.14 (Which examples enter the GW theorem). Every finite example above satisfies [corollary 4.5](#). The finite-alphabet sequence spaces, Cantor set, $\mathbb{Z}_p$, compact unit balls such as $\mathbb{F}_q[[t]]$, profinite groups, bounded-degree rooted graph space, compact Hausdorff hyperspaces and locally finite tree boundaries all satisfy the hypotheses of [theorem 6.1](#). Noncompact examples such as Baire space, $\mathbb{Q}_p$, full Laurent-series fields and infinite-dimensional sequence spaces require either compact restrictions or an extension with moment and integrability assumptions. Ordinary shortest-path tree metrics do not enter this list unless their restriction has separately been shown to be ultrametric.

6 Compact ultrametric spaces

Ultrametric geometry supplies a concrete and useful class where distance thresholds are automatically conditionally negative. At each scale, the space breaks into disjoint clusters, turning the key product quadratic form into a sum of squares.

Theorem 6.1 (Conditional concavity on compact ultrametric spaces). *Let (X, d_X) and (Y, d_Y) be compact ultrametric spaces. For arbitrary probability measures $\alpha \in \mathcal{P}(X)$ and $\beta \in \mathcal{P}(Y)$, and every submodular mixed-measure distortion ℓ , the objective*

$$\pi \mapsto \iint \ell(d_X(x, x'), d_Y(y, y')) d\pi(x, y) d\pi(x', y')$$

is concave on $\Pi(\alpha, \beta)$. In particular, this holds for every power distortion $\ell(a, b) = |a - b|^p$, $p \geq 1$.

Proof. Fix $r > 0$. In an ultrametric space, the relation $x \sim_r x'$ if $d_X(x, x') < r$ is an equivalence relation. Compactness implies that its partition $\mathcal{P}_r^X$ has finitely many classes. Therefore

$$H_r^{d_X}(x, x') = 1 - \sum_{C \in \mathcal{P}_r^X} \mathbf{1}_C(x) \mathbf{1}_C(x').$$

For every finite signed measure σ of total mass zero,

$$\iint H_r^{d_X}(x, x') d\sigma(x) d\sigma(x') = - \sum_{C \in \mathcal{P}_r^X} \sigma(C)^2 \leq 0.$$

Thus every threshold kernel is CND, in agreement with [proposition 4.2](#), and the same argument applies on Y .

More directly, if $\xi \in \mathcal{T}_{X,Y}$, then the analogue of (6) is

$$I_\xi(r, s) = \sum_{C \in \mathcal{P}_r^X} \sum_{D \in \mathcal{P}_s^Y} \xi(C \times D)^2 \geq 0. \quad (25)$$

Indeed, the constant term and the terms involving only one partition vanish because ξ has zero total mass and zero marginals. The same Fubini calculation as in (9) gives

$$\iint \ell(d_X(x, x'), d_Y(y, y')) d\xi(x, y) d\xi(x', y') = - \int I_\xi(r, s) d\omega_\ell(r, s) \leq 0.$$

The quadratic segment expansion used in [proposition 1.1](#) therefore proves concavity. The power case follows from [corollary 4.4](#). $\square$

Remark 6.2 (The endpoint $p = \infty$). The usual $p = \infty$ GW distortion is an essential supremum rather than a quadratic functional of the coupling. The direction-space notion of conditional concavity used here therefore does not pass to that endpoint.

7 Conclusion and recommended book statement

The clean presentation should begin with the master identity (9), not with two apparently unrelated proofs. For genuine metrics, it yields a hierarchy of certificates:

$$\text{ultrametric} \iff \text{threshold-CND} \implies \text{CND}.$$

At $p = 2$, the constant scale weight allows one to use only the weakest, integrated CND/CPD certificate, as formalized in [corollary 2.3](#). Under the stronger ultrametric certificate, the same formula controls every pair of scales and therefore every $p \geq 1$, as well as every submodular mixed-measure distortion. Thus the ultrametric theorem genuinely contains the quadratic case; what is special about $p = 2$ is that its hypothesis can be weakened.

The counterexamples in [theorem 3.1](#) show that this weakening cannot simply be reused for $p \neq 2$. In finite ultrametric spaces, [corollary 4.5](#) additionally yields sparse and, for uniform marginals, permutation optimizers. This organization also prevents confusing the outer GW exponent with powers used inside the base dissimilarities.

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